Basic Statistical Concepts
Part II

Looking at relationship

Two variables can be related in three ways:

(1) Positively related: If one variable increases other also increases or one variable decreases other also decreases

(2) Not related: There is no relationship between the two variables such that the value of one variable changes and the other variable remains constant

(3) Negatively related: If one variable increases other decreases or one variable decreases other increases

Looking at relationship

How we can express the relationship between variables statistically by looking at two measures:

- Covariance
- Correlation coefficient

How do we measure relationship?

The simplest way to look at whether two variables are associated is to look a whether they <u>covary</u>

To understand what <u>covariance</u> is, we first need to think back to the concept of variance

The variance of a single variable represents the average amount that the data vary from the mean

Numerically, it is described by:

$$\text{Variance } (SD^2) = \frac{\sum (X - M)^2}{N} = \frac{\sum (X - M)(X - M)}{N}$$

How do we measure relationship?

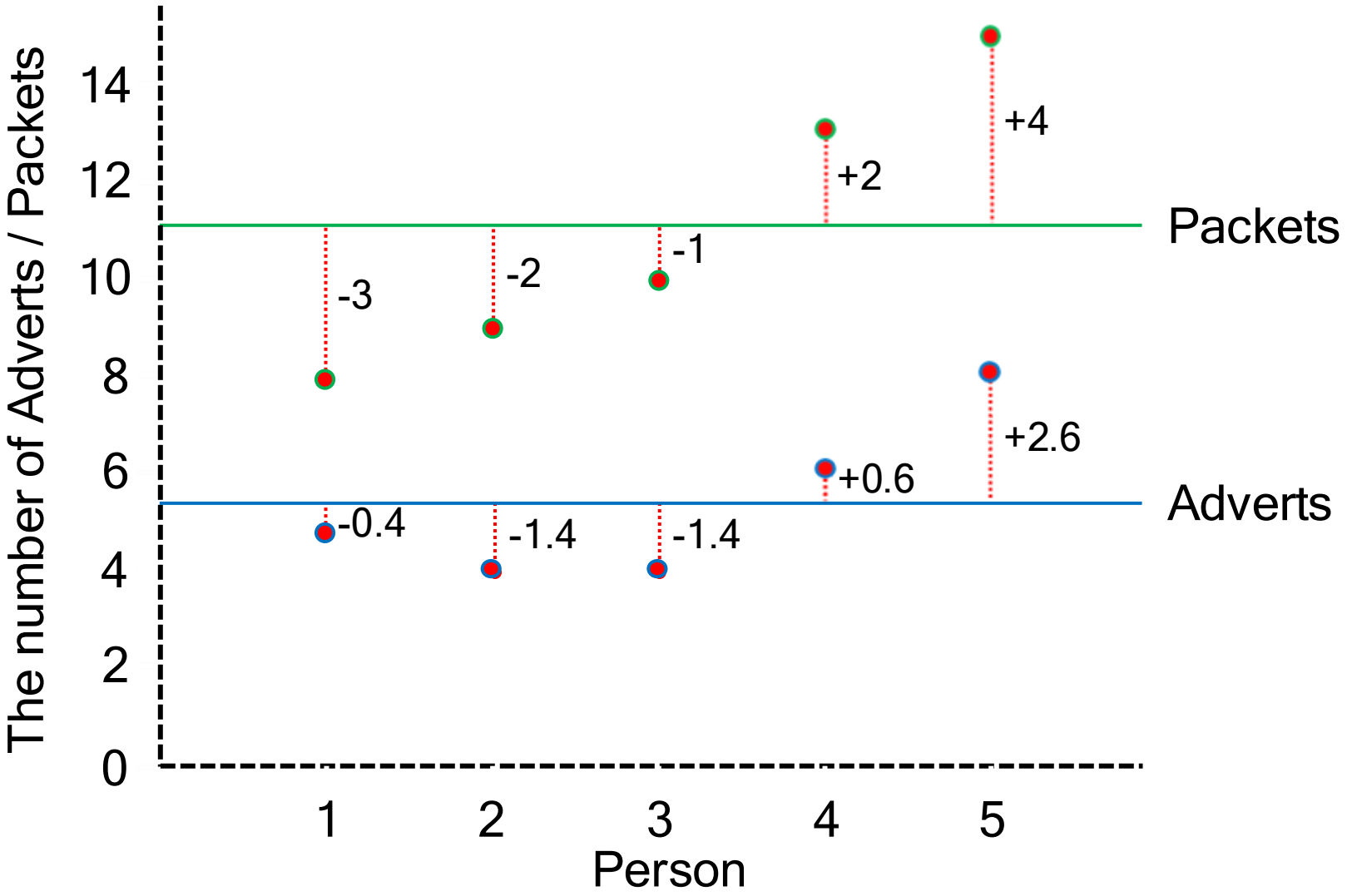
If we are interested in whether two variables are related, then we are interested in whether changes in one variable are met with similar changes in the other variable

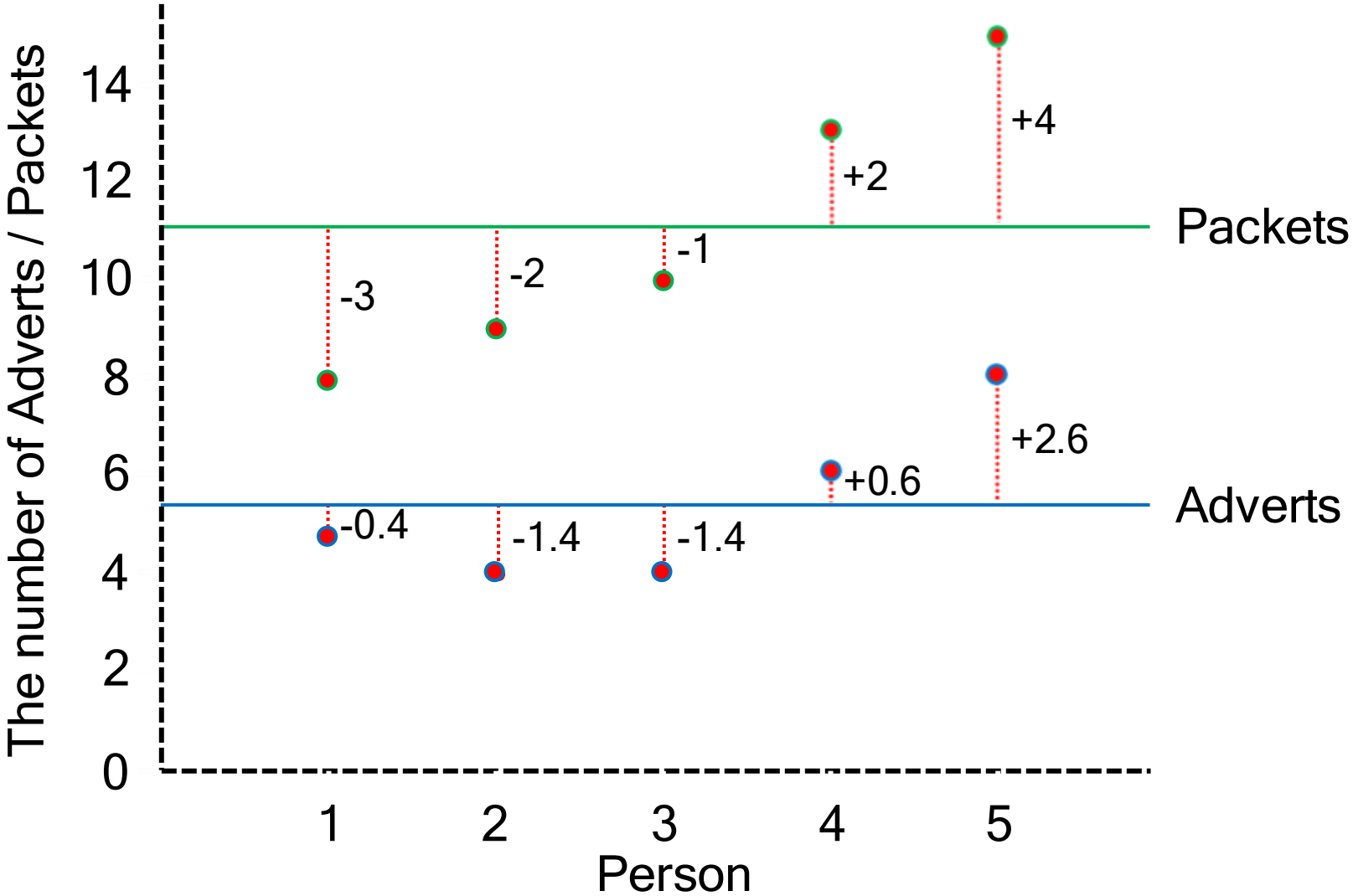
Therefore, when one variable deviates from its mean we would expect the other variable to deviate from its mean in a similar way

To illustrate, imagine we took five people and subjected them to a certain number of advertisements promoting toffee sweets

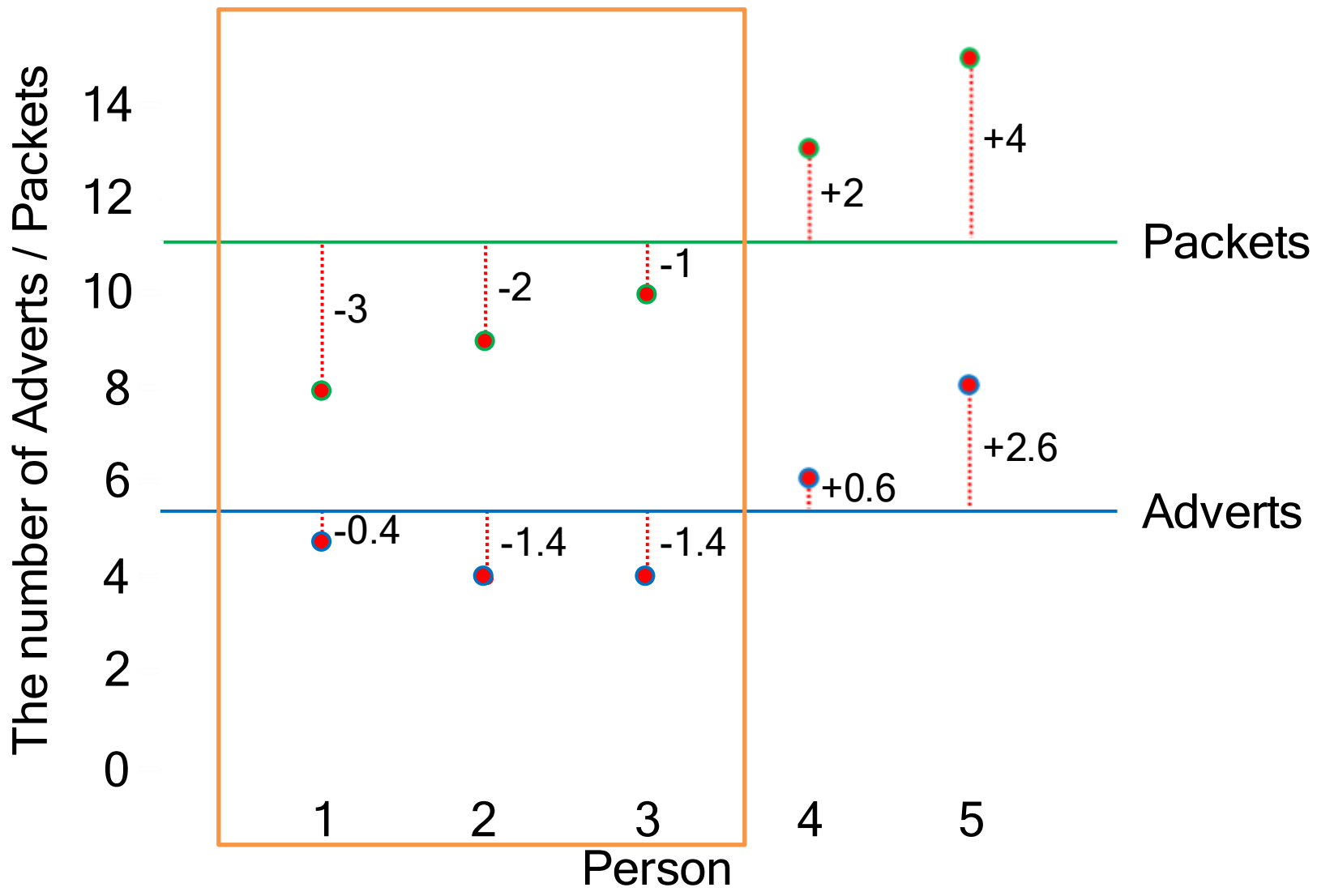
And then measured how many packets of those sweets each person bought during the next week

Subject	1	2	3	4	5	Mean	SD
Adverts Watched	5	4	4	6	8	**5.4**	**1.67**
Packets Bought	8	9	10	13	15	**11.0**	**2.92**

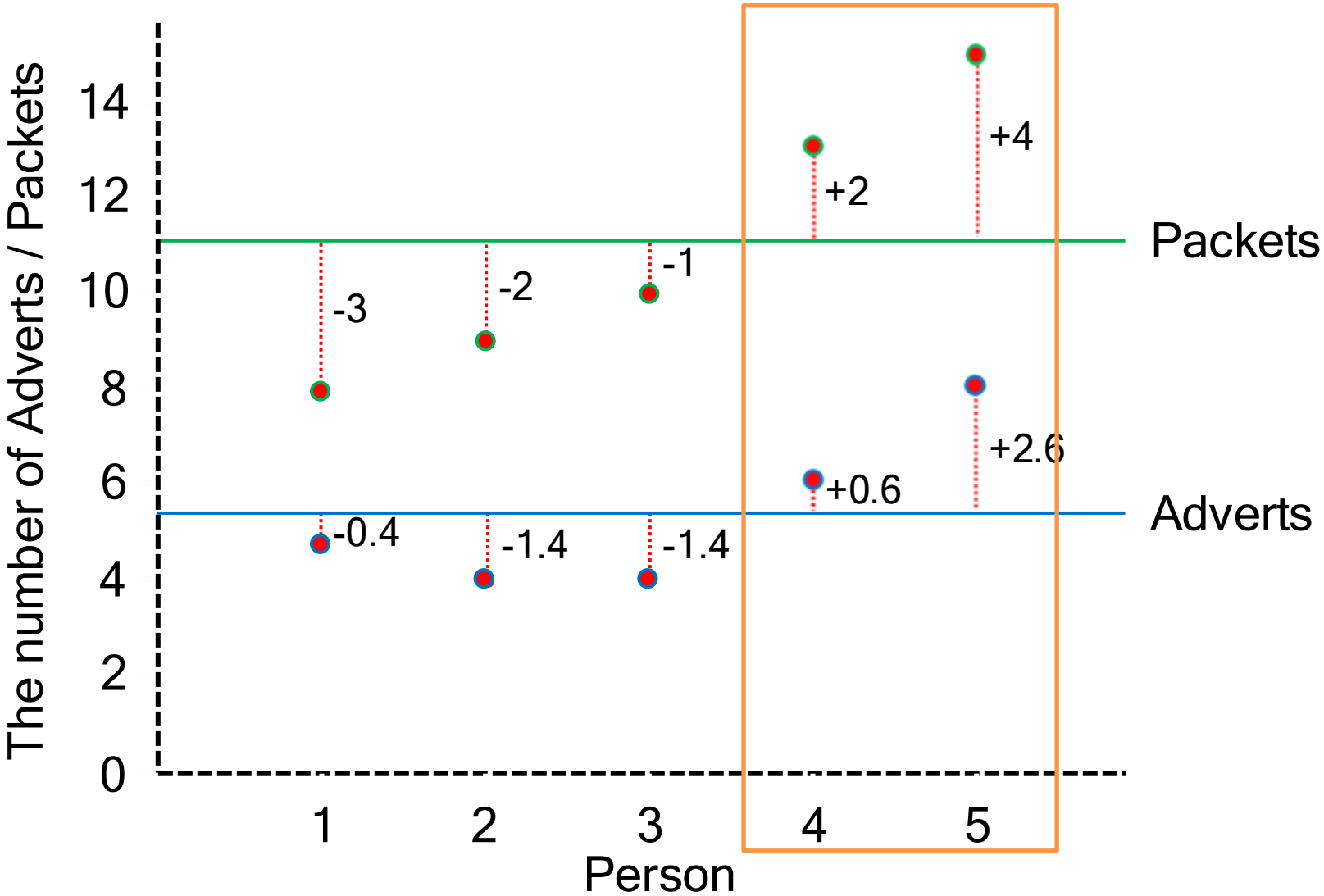




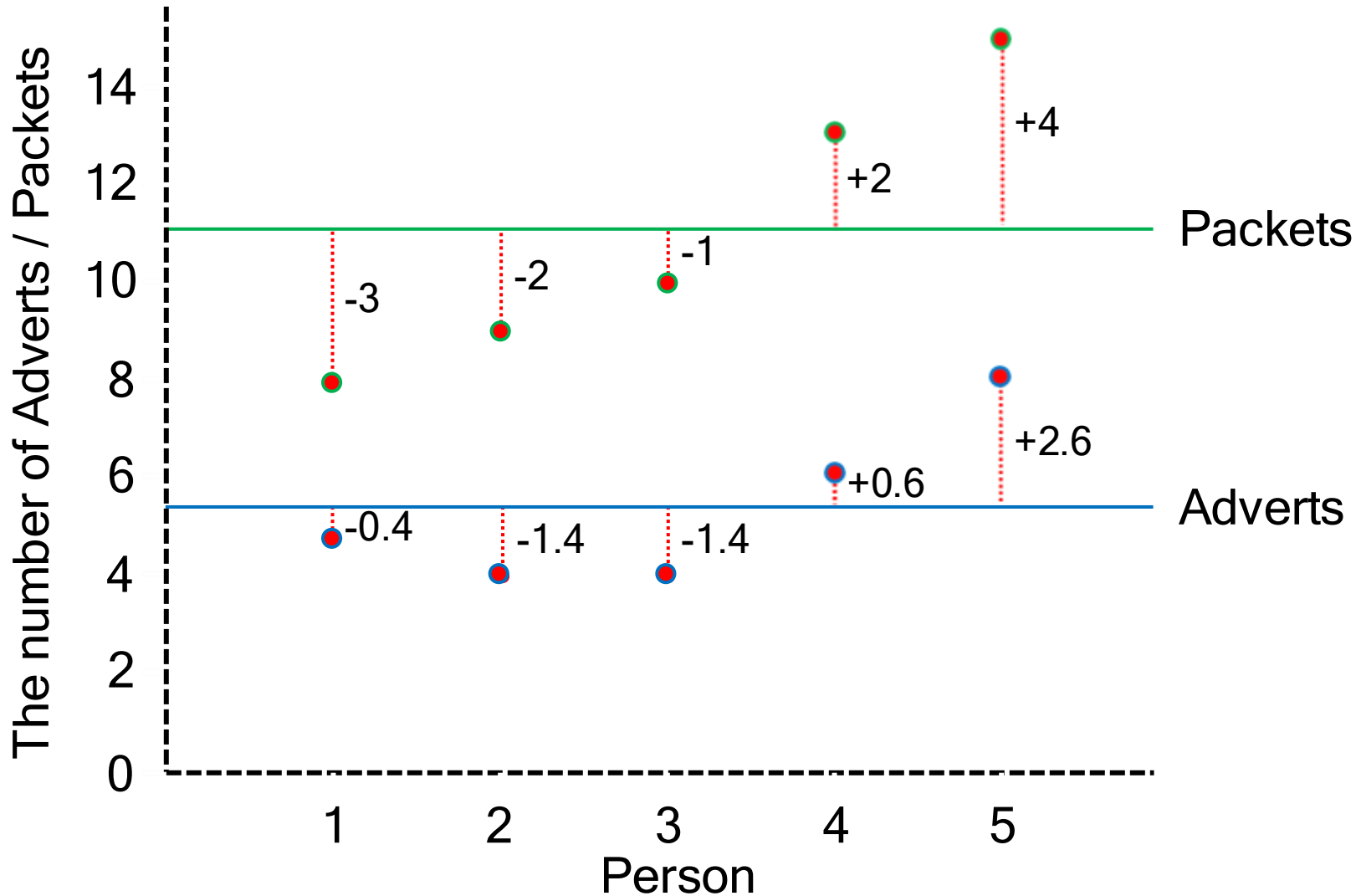
The first think to notice about this deviation graph is that there is a very similar pattern of deviations for both variables



For the first three participants the obsereved values are below the mean for both variables



For the last two participants the obsereved values are above the mean for both variables



This pattern is indicative of a potential relationship between the two variables

Because it seems that if a person's score is below the mean for one variable then their score for the other will also be below the mean

So, how do we calculate the exact similarity between the pattern of differences of two variables?

To calculate this relationship, we will use the variance formula by making some modifications

$$\text{Variance } (SD^2) = \frac{\sum (X - M)^2}{N} = \frac{\sum (X - M)(X - M)}{N}$$

$$\text{Covariance} = \frac{\sum (X - Mx)(Y - My)}{N - 1}$$

X is a score of the first variable

Mx is mean of the first variable

Y is a score of the second variable

My is mean of the second variable

Person	X (Adverts)	Mx	X - Mx	Y (Packets)	My	Y - My	Cross-Product Deviation
1	5	5.4	-0.4	8	11	-3	1.2
2	4	5.4	-1.4	9	11	-2	2.8
3	4	5.4	-1.4	10	11	-1	1.4
4	6	5.4	+0.6	13	11	+2	1.2
5	8	5.4	+2.6	15	11	+4	10.4

Person	X (Adverts)	Mx	X - Mx	Y (Packets)	My	Y - My	Cross-Product Deviation
1	5	5.4	-0.4	8	11	-3	1.2
2	4	5.4	-1.4	9	11	-2	2.8
3	4	5.4	-1.4	10	11	-1	1.4
4	6	5.4	+0.6	13	11	+2	1.2
5	8	5.4	+2.6	15	11	+4	10.4
						∑ =	17

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4	6	5.4	+0.6	13	11	+2	1.2
5	8	5.4	+2.6	15	11	+4	10.4
						∑ =	17

Calculating the covariance is a good way to assess whether two variables are related each other

A positive covariance indicates that as one variable deviates from the mean, the other variable deviates in the same direction

On the other hand, a negative covariance indicates that as one variable deviates from the mean (e.g. increases), the other deviates from the mean in the opposite direction (e.g. decreases)

However, there is one problem with covariance as a measure of the relationship between variables

It depends upon the scales of measurement used

So, covariance is not a standardized measure

For example, if we use the previuos data and assume that they represented two variables measured in miles then the covariance is 4.25

If we then convert these data into kilometers and calculate the covariance again then we should find that it increases to 11

This dependence on the scale of measurement is a problem because it means that we cannot compare covariances in a objective way

Standardization of covariance: The correlation coefficient

To overcome the problem of dependence on the measurement scale, we need to convert the covariance into a standard set of units

This process is known as <u>standardization</u>

$$\text{Cov}(X, Y) = \frac{\sum (X - Mx)(Y - My)}{N - 1}$$

$$r = \frac{\text{Cov}(X, Y)}{SDx * SDy}$$

Cov (X, Y) = 4.25

SDx = 1.67

SDy = 2.92

r = 4.25 / (1.67 * 2.92)

r = 4.25/ 4.88

r = .87

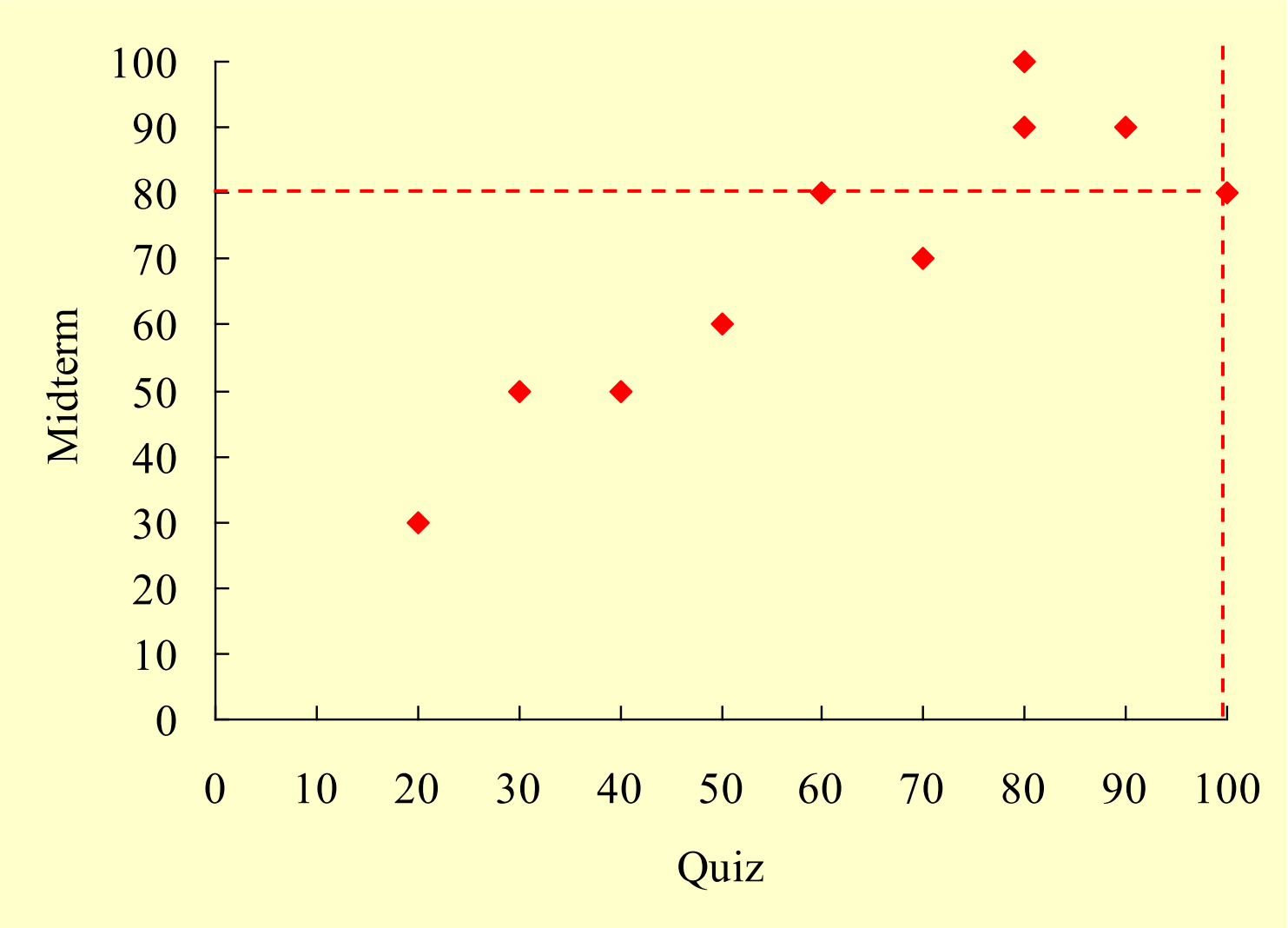
This standardized covariance is known as Pearson correlation coefficient

This coefficient not only enables us to determine *the direction of the relationship*; but also enables us to determine *the strength or magnitude of the relationship* between the two variables

Perfect	+1									-1
		+.90							-.90	
Strong		+.80							-.80	
		+.70							-.70	
			+.60					-.60		
Moderate			+.50					-.50		
			+.40					-.40		
				+.30			-.30			
Weak				+.20			-.20			
				+.10			-.10			
Zero						0				

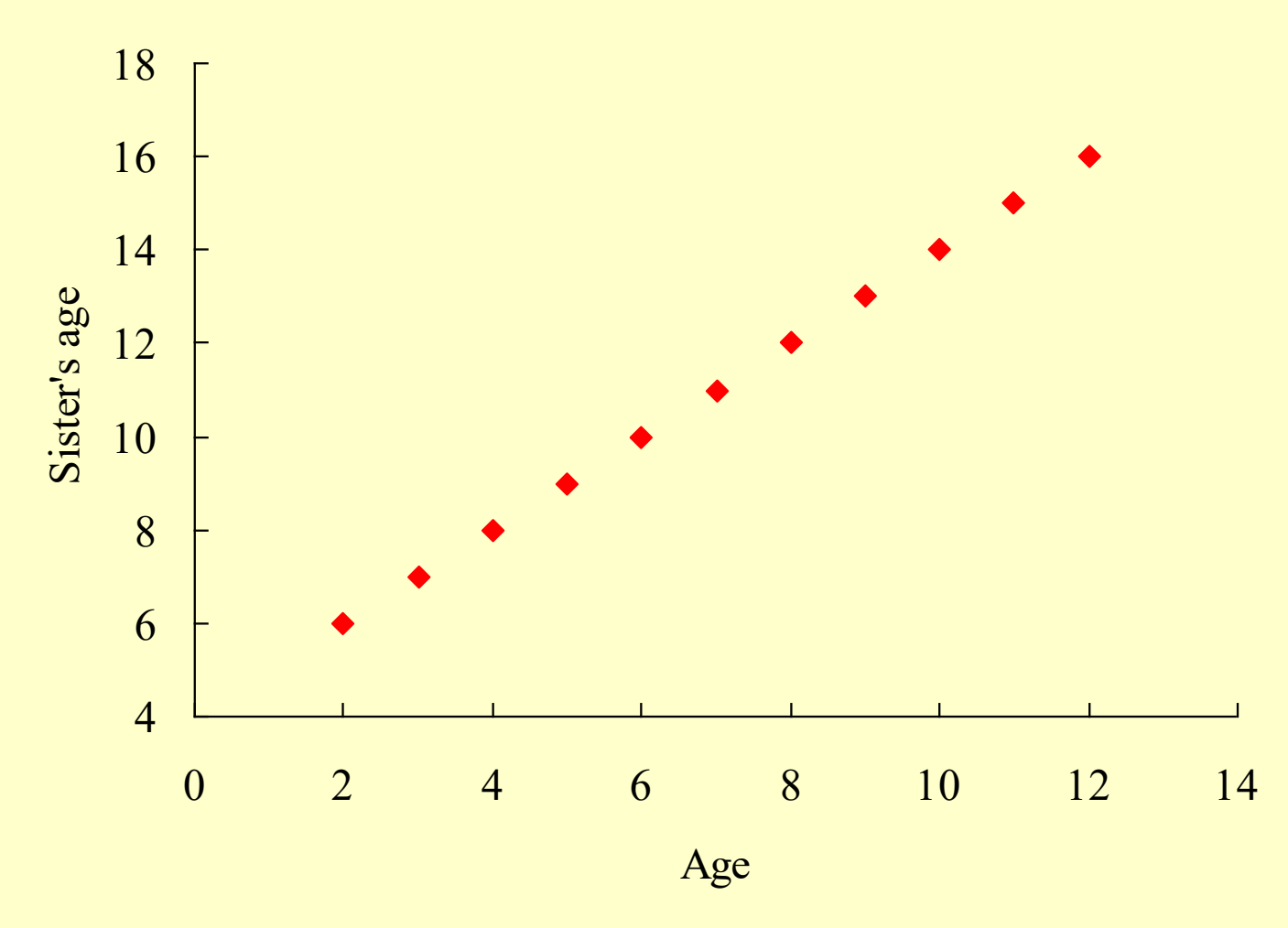
Scatterplot

QUIZ	MIDTERM
50	60
70	70
30	50
80	90
20	30
60	80
40	50
90	90
80	100
100	***80***



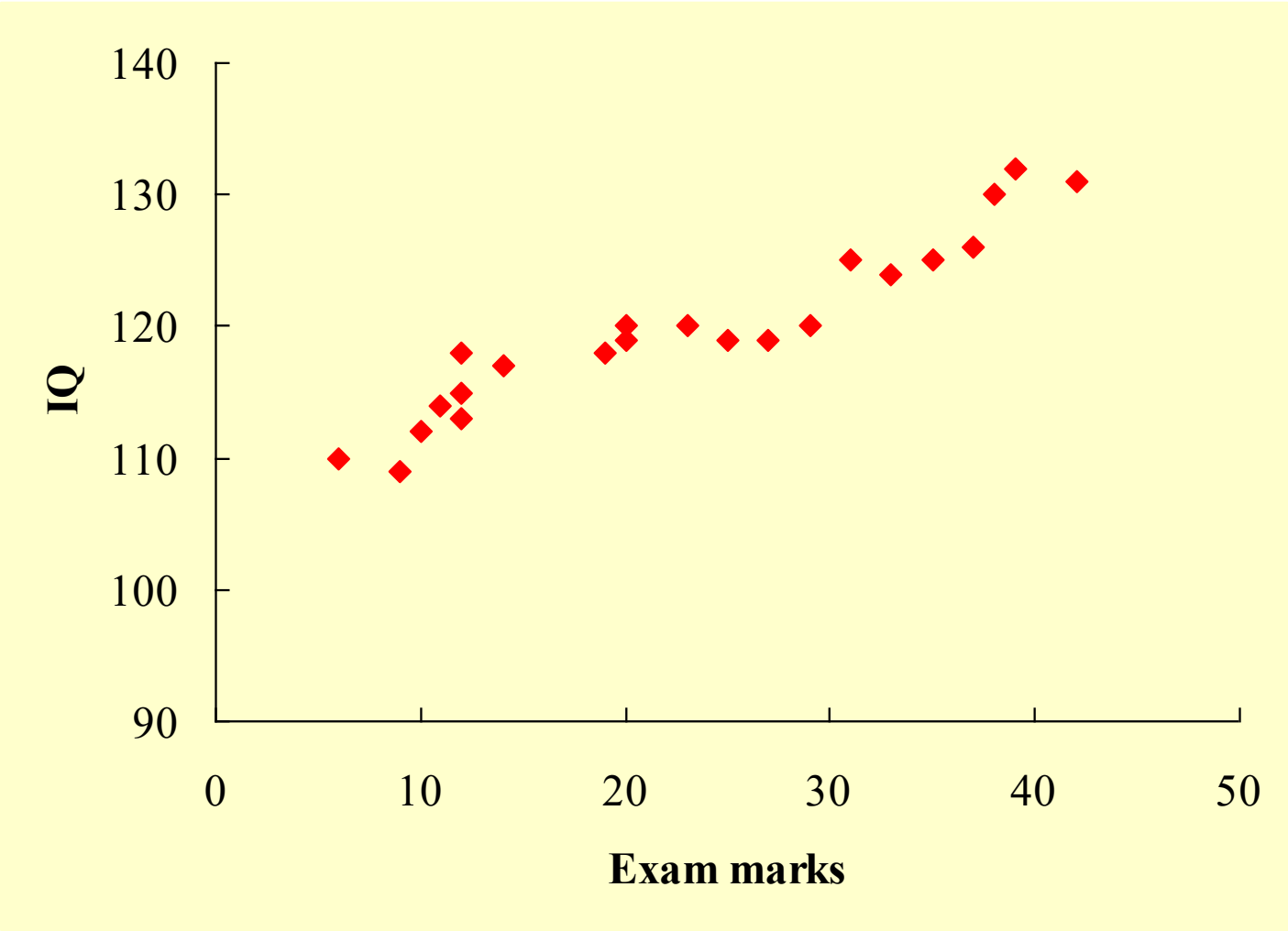
Direction of the relationship

Perfect positive relationship



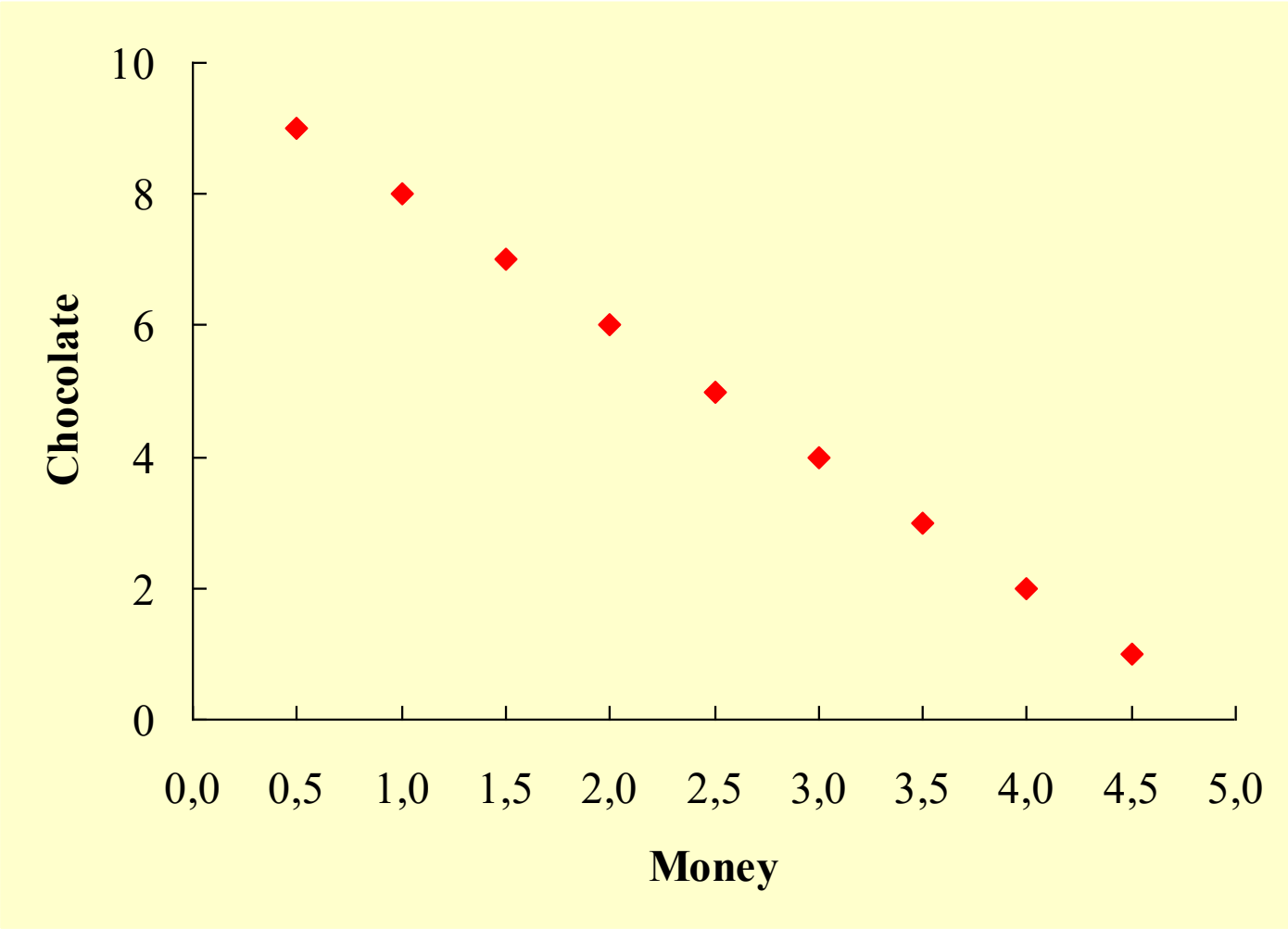
Direction of the relationship

Imperfect positive relationship



Direction of the relationship

Perfect negative relationship



Bivariate correlation

- Some other types of bivariate correlations

  1. Pearson correlations (i.e., special cases of r_{XY}):

Coefficient	Estimates the association between
Point-biserial (r_{pb})	a dichotomous variable and a continuous variable
Phi ($\hat{\varphi}$)	two dichotomous variables
Spearman's rank order (rho)	two ordinal variables

Bivariate correlation

- Some other types of bivariate correlations

  2. Non-Pearson correlations that estimate r_{XY} assuming both variables were continuous and normally distributed

Coefficient	Applied to
Biserial	a dichotomous variable and a continuous variable
Polyserial	a categorical variable with two or more levels and a continuous one
Tetrachoric	two dichotomous variables
Polychoric	two categorical variables each with two or more levels

Bivariate correlation and regression

- The linear regression equation for unstandardized variables is:

$$\hat{Y} = B\,X + A$$

$B = r_{XY}\,(SD_Y / SD_X)$ is the *unstandardized regression coefficient* (i.e., the slope)

$A = M_Y - B\,M_X$ is the *intercept (constant)*

Bivariate correlation and regression

- The value of *B* is the predicted difference on *Y*, given a difference of 1 point on *X*
- The value of *A* equals $\hat{Y}$, given $X = 0$
- The values of *B* and *A* in a particular sample are those that minimize the sum of the squared residuals, $\Sigma\ (Y - \hat{Y})^2$
- That is, they satisfy the *least squares criterion*

Bivariate correlation and regression

- Example:

$$\hat{Y} = 2.30\,X + 10.00$$

A 1-point difference on X predicts the same on Y of 2.30 points

$\hat{Y} = 10.00$, given $X = 0$

- Because B is an unstandardized estimate, its values are not limited to a particular range

- Consequently, a large numerical value of B does not necessarily mean that X is an “important” or “strong” predictor of Y

How to measure relationships between two variables

These correlations can be very useful but we can take this process a step further and predict one variable from another

A simple example might be to try to predict levels of stress from the amount of time until you have to give a talk

You'd expect this to be a negative relationship: The smaller the amount of time until the talk, the larger the anxiety

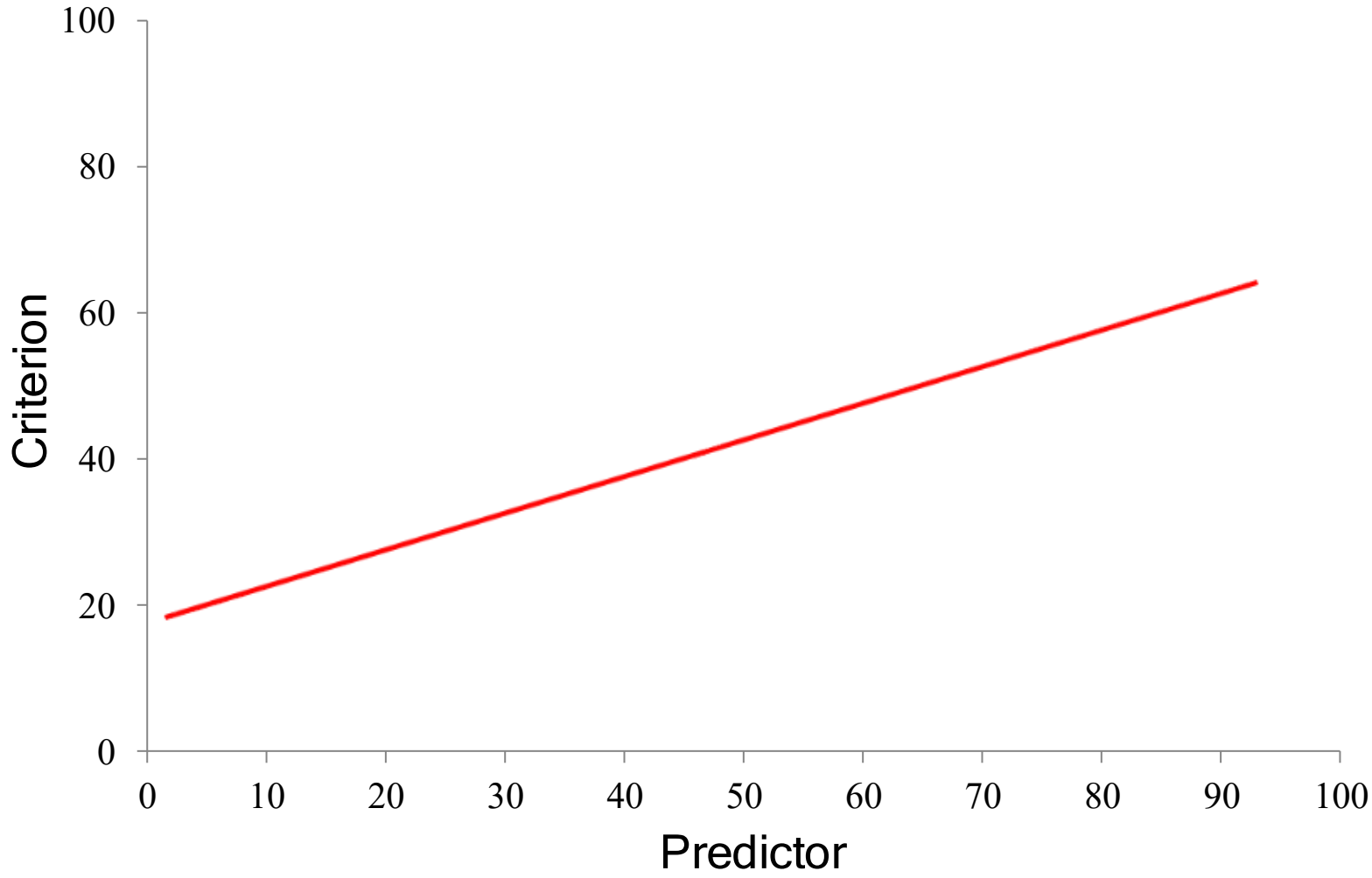
We could then extend this basic relationship to answer a question such as: *If there's 10 minutes to go until someone has to give a talk, how anxious will they be?*

This is the essence of regression analysis:

We fit a model to our data and use it to predict values of the dependent variable from the independent variable

Regression analysis is a way of predicting a **criterion** (dependent variable) variable from **predictor** (independent variable) variable

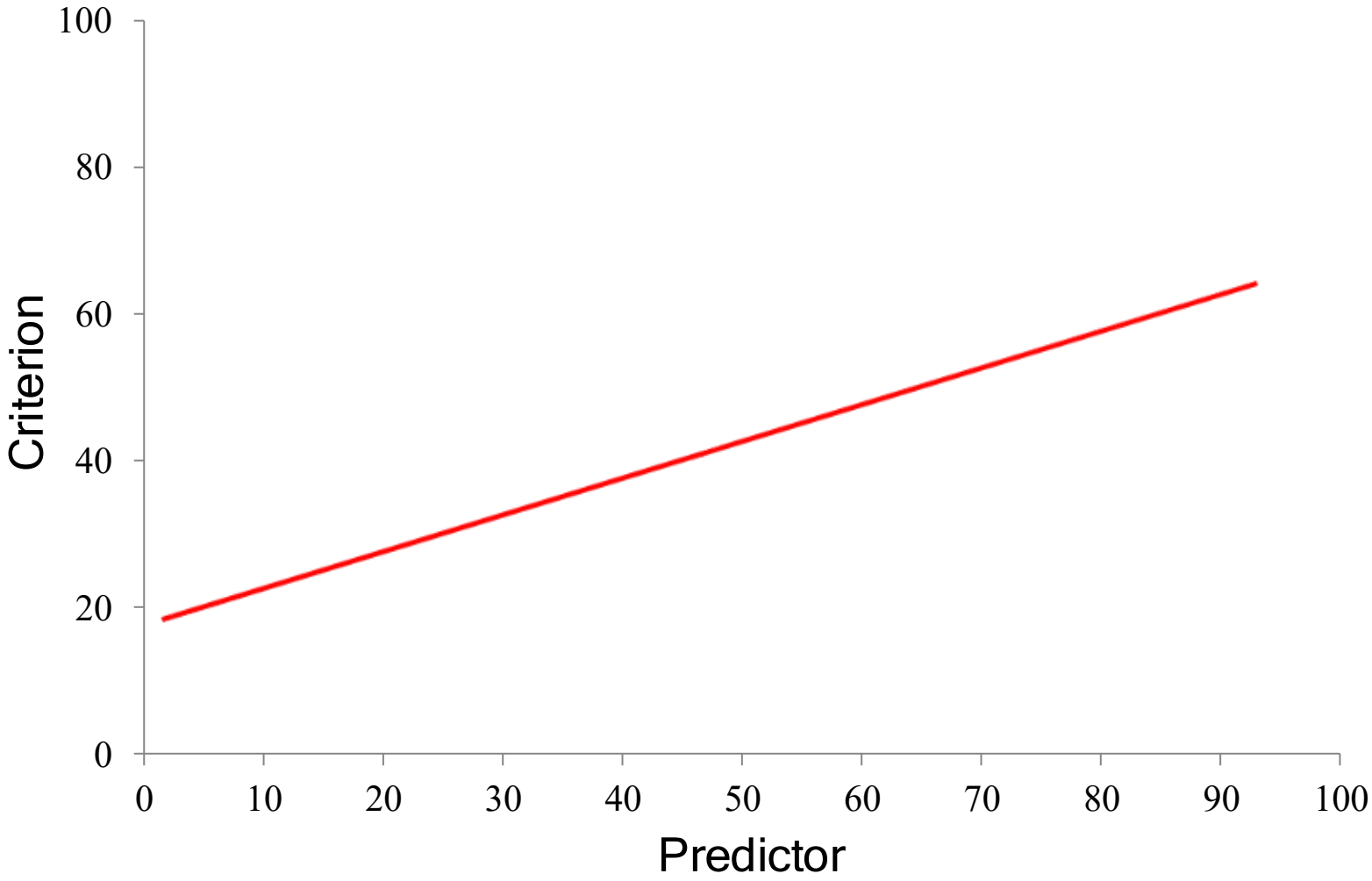
The simplest way to do this would be find a line that best describes the data



Any straight line can be defined by two things:

1. The slope of the line (usually denoted by *b*)

2. The intercept (usually denoted by *a*): the point at which the line crosses the vertical axis of the graph



We can represent this line with an equation:

$$Y = a + bX + \varepsilon$$

Y; criterion variable

a; intercept, *b*; slope

X; predictor variable

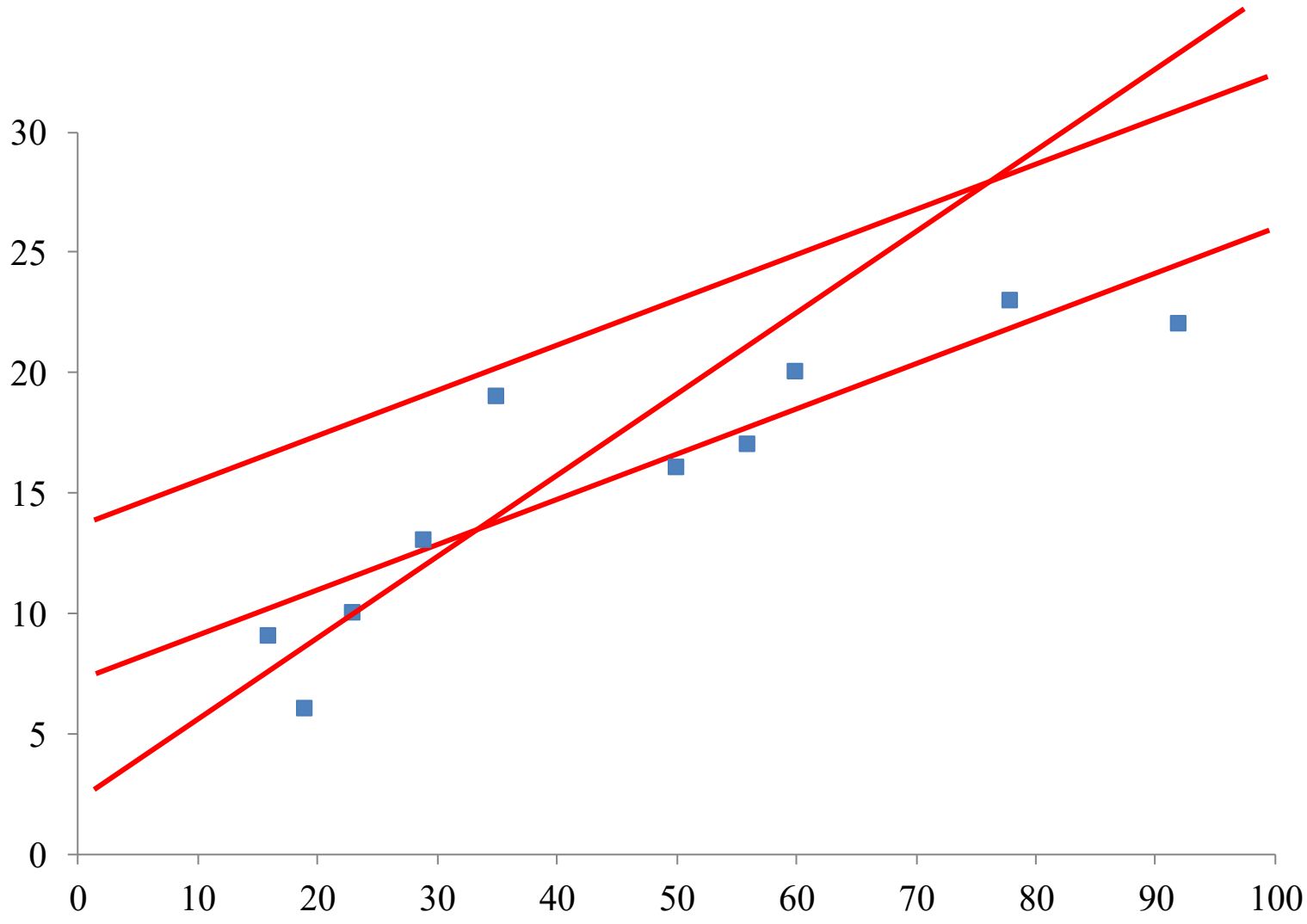
ε; residual

How to find the line that best fits the data?

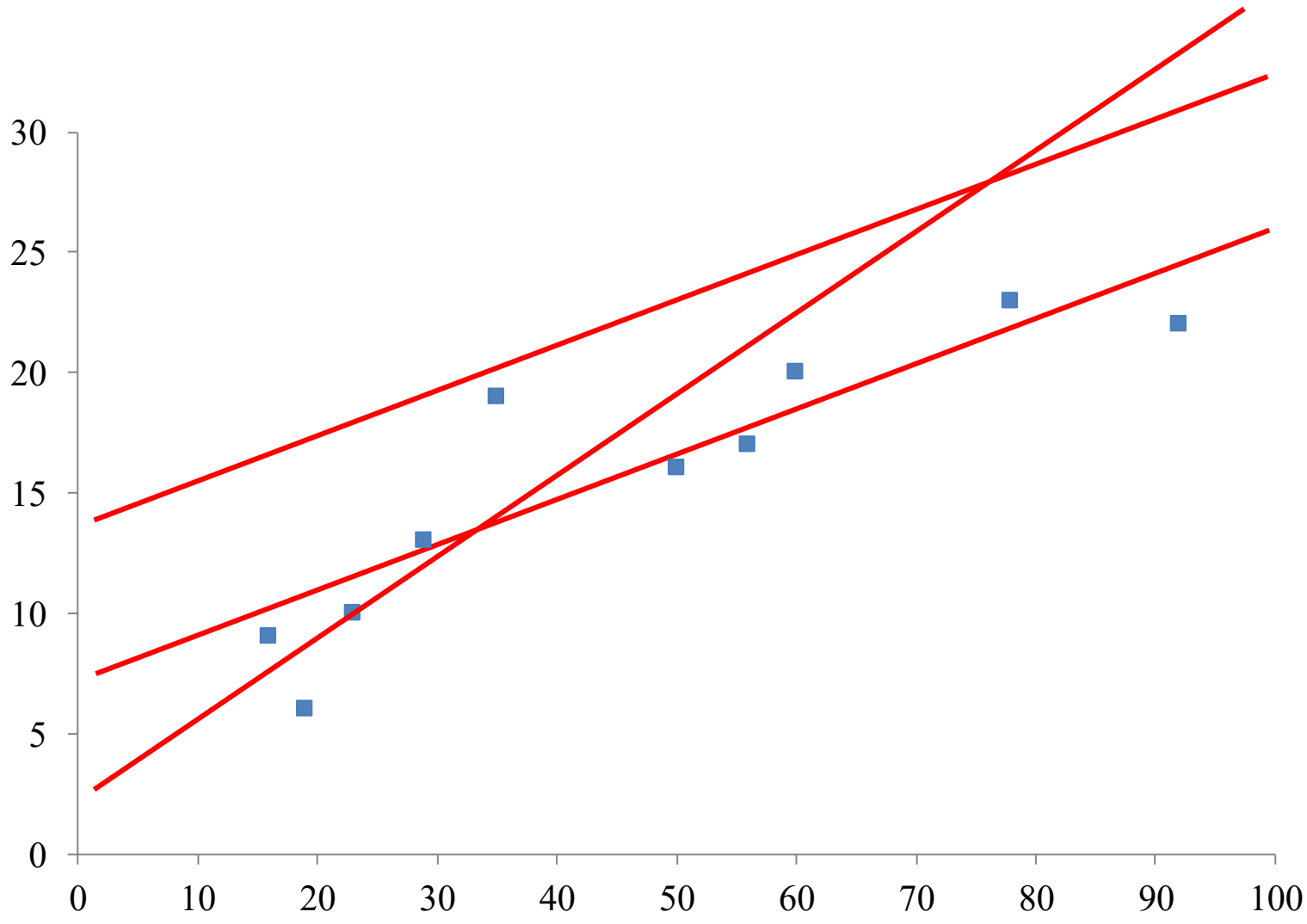
In other words, how to find a line that goes through, or as close to, as many of the data points as possible?

This “line of best fit” is found by using <u>the method of least squares</u>

Some possible lines can be drawn



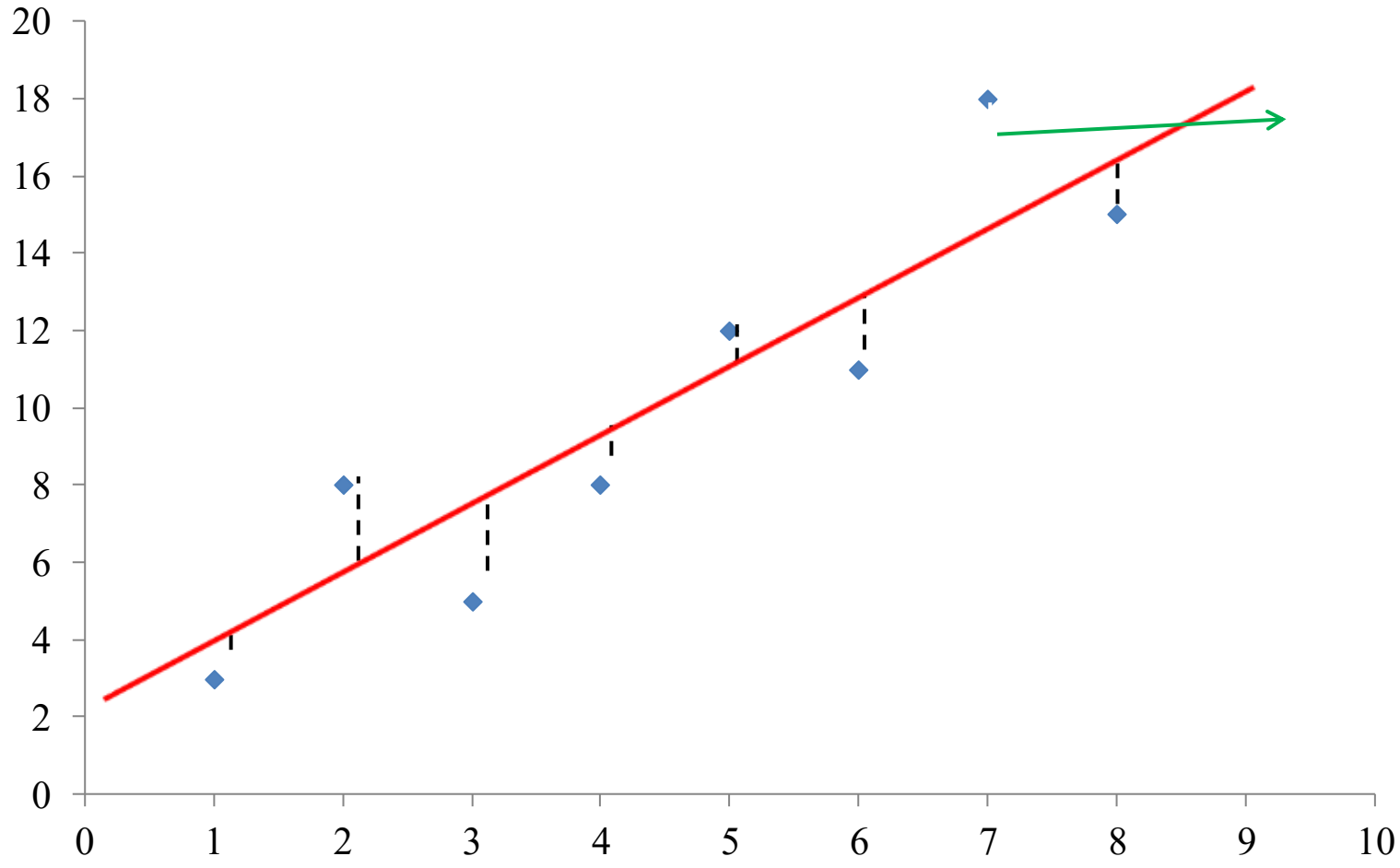
The aim of the method of least squares is to decide which line results in the least amount of difference between the observed data points and the line



In the method of least square, first it is calculated the difference between data points and the regression line for all possible lines

Sum of squared residuals

The lowest SS_R is the line of best fit



We assess **the goodness of fit** the model

We do this because even though this line is the best one available, it can still be a lousy fit to the data!

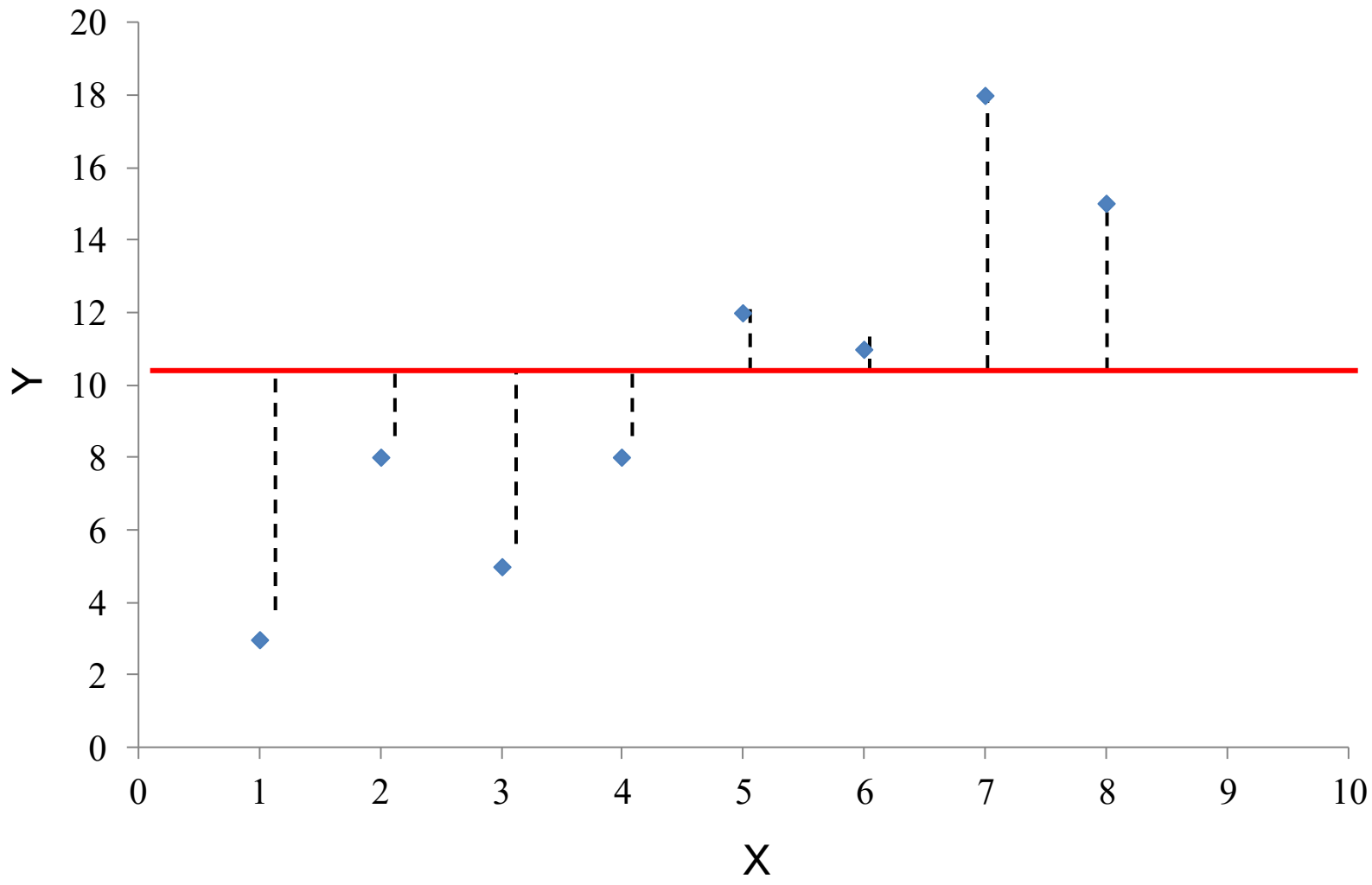
For assessing the goodness of fit, we need three “Sum of Squares” values

- Total Sum of Squares

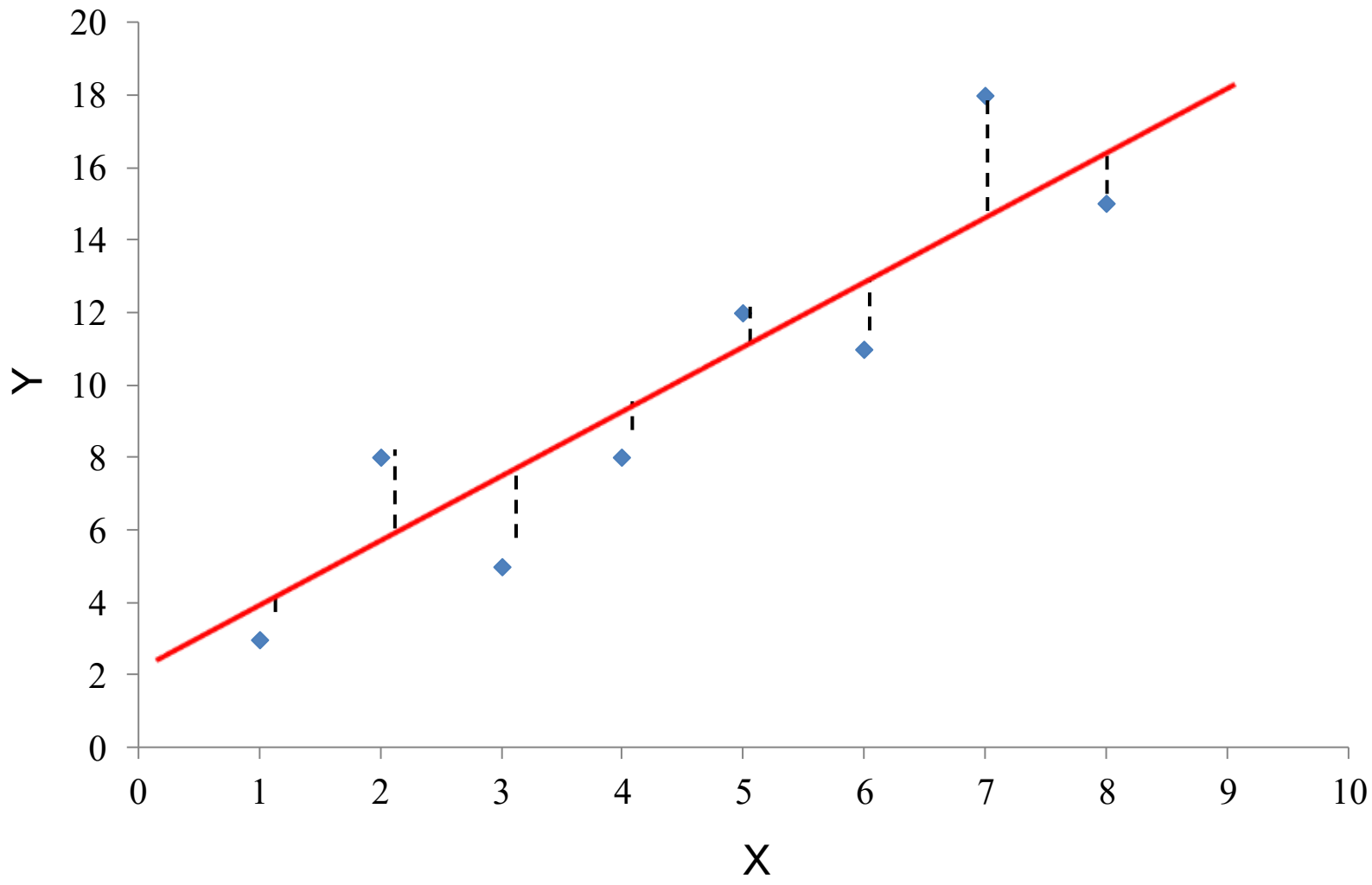
- Residual Sum of Squares

- Model Sum of Squares

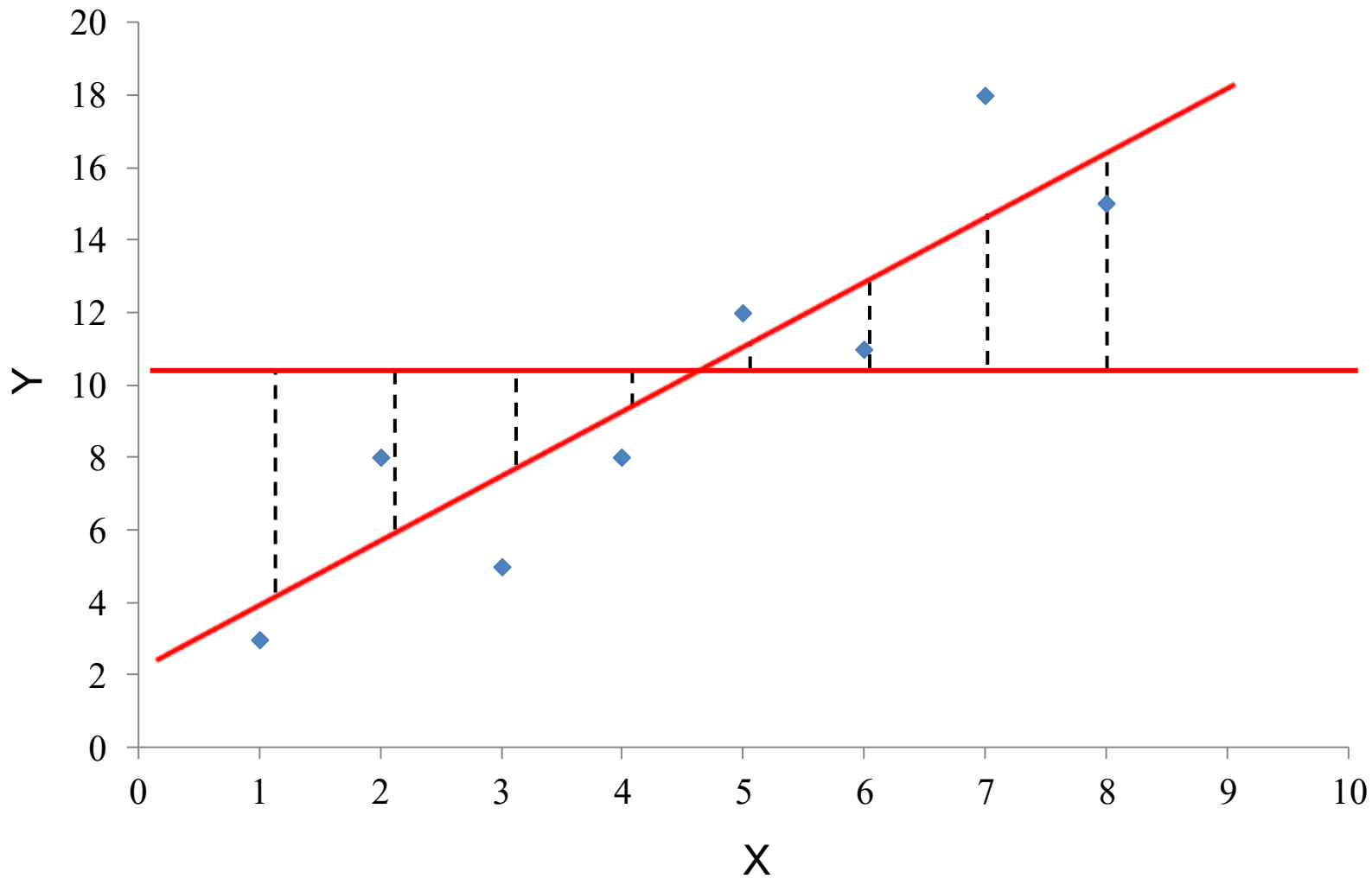
SS_T uses the differences between the observed data and the mean value of Y



SS_R uses the differences between the observed data and the regression line



SS_M uses the differences between the mean value of Y and the regression line



If the value of SS_M is large then the regression model is very different from using the mean predict the criterion variable

However, if the value of SS_M is small then using the regression model is little better than using the mean

This is calculated by dividing the sum of the squares for the model by the total sum of squares

$R^2 = SS_M / SS_T$

A second use of the sums of squares in assessing the model fit is through the F-test:

$F = MS_M / MS_R$

$MS_M = SS_M / df_M$

$MS_R = SS_R / df_R$

df_M = the number of variable – 1

df_R = the number of score – 2

Why This Matters for Psychometrics

Correlation is the foundation of:

- Reliability
- Validity
- Factor analysis
- SEM

Regression logic underlies:

- Prediction
- Criterion validity